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AUTONOMOUS MACHINE COMPUTING

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Autonomous Machine Computing

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Executive Summary

The field of Embodied AI marks a significant evolution in how AI and robotics intersect to enable autonomous, adaptive, and intelligent systems. By embedding cognitive capabilities within physical entities, Embodied AI equips machines to interact meaningfully with their environments, merging perception, decision-making, and action in a seamless loop. This 2024 update to the Autonomous Machine Computing White Paper under IEEE IRDS reflects on the transformative milestones that have shaped Embodied AI, from the early conceptual foundations of machine perception to today's sophisticated multimodal learning models. As Embodied AI continues to integrate advances in deep learning, autonomous machine computing, and sensory data processing, it holds the potential to redefine industries and elevate human-machine collaboration. This paper not only explores the current challenges and innovations in Embodied AI—from foundational model-building and specialized computing infrastructures to the pivotal role of data—but also addresses the societal, ethical, and regulatory frameworks needed to guide Embodied AI's responsible integration into our lives.

1. Introduction to Embodied AI

Embodied AI signifies a transformative convergence of AI and robotics, embedding intelligence directly into physical forms like robots and intelligent machines, which enables these systems to engage in meaningful interactions with their environments [1]. Unlike conventional AI, which operates primarily in abstract data spaces, Embodied AI grounds itself in real-world interactions, seamlessly integrating perception, cognition, and action. This section traces the foundational milestones that have driven Embodied AI's progress and offers insights into its future trajectory.

The concept of Embodied AI first gained significant attention in Rodney Brooks's 1991 paper, "Intelligence without Representation." [12] Brooks proposed a groundbreaking perspective that intelligent behavior could emerge from an autonomous machine's physical interactions with its surroundings, challenging the notion that complex algorithms or internal representations are necessary for intelligence.

In 1999, Rolf Pfeifer and Christian Scheier advanced this idea in their book, "Understanding Intelligence," arguing that intelligence transcends the brain or specific algorithms, manifesting through the entire physical structure and function of an agent [13]. They contended that intelligence is fundamentally shaped by the body itself, opposing traditional brain- and computation-centric views and emphasizing the body's central role in shaping intelligent behavior.

Building on insights from cognitive science, Linda Smith introduced the "embodiment hypothesis" in 2005, which highlights the crucial role of the body's interactions with the environment in forming cognitive processes [14]. According to this hypothesis, our cognition, perception, and abilities are products of ongoing interactions between our physical selves and the external world. The environment not only provides sensory inputs but actively participates in shaping our behaviors and cognitive frameworks.

These foundational studies establish three key principles for developing Embodied AI systems. First, such systems should not rely on predefined, intricate logic to handle specific scenarios. Second, Embodied AI systems should integrate evolutionary learning mechanisms, allowing them to adapt dynamically to their environments. Lastly, the environment plays a critical role, influencing not only physical actions but also cognitive development.

Looking ahead, Embodied AI is set to harness deep learning, reinforcement learning, and multimodal AI models. By integrating these technologies, future robots and AI systems could evolve beyond task-specific programming to achieve flexible, generalizable learning. As these capabilities advance, Embodied AI is anticipated to transform various industries, redefine human-machine interactions, and deliver far-reaching societal benefits.

2. Building Foundation Models for Embodied AI

A critical component in the development of advanced Embodied AI systems is the creation of foundational models that can generalize across a wide range of tasks and environments. These foundational models, which are designed to learn adaptive strategies rather than isolated skills, form the core of effective Embodied AI systems [2]. Drawing inspiration from human cognitive abilities, these models empower machines to perform context-sensitive tasks without the need for exhaustive, task-specific programming.

A prevalent approach for constructing these foundational models is to utilize large pretrained models, such as GPT, as a baseline. These models are often fine-tuned and enhanced with techniques like in-context learning (ICL) to improve their performance. Pretrained models typically contain a vast number of parameters that encode a wide range of world knowledge, and they offer fast response times due to their relatively small context windows. This pre-encoding enables these models to exhibit strong zero-shot performance. However, the limited context windows present challenges for continuous learning from the operating environment of the Embodied AI system, as well as for establishing connections across diverse usage scenarios.

Alternatively, another approach focuses on models with fewer parameters but larger context windows. These models, which do not attempt to encode the entirety of world knowledge, instead concentrate on meta-learning—learning how to learn. By utilizing larger context windows, these models can engage in general-purpose in-context learning (GPICL), enabling them to continuously adapt to their environment and establish meaningful connections across a broad spectrum of contexts. This approach facilitates more dynamic and scalable learning, making it well-suited for the evolving needs of Embodied AI systems.

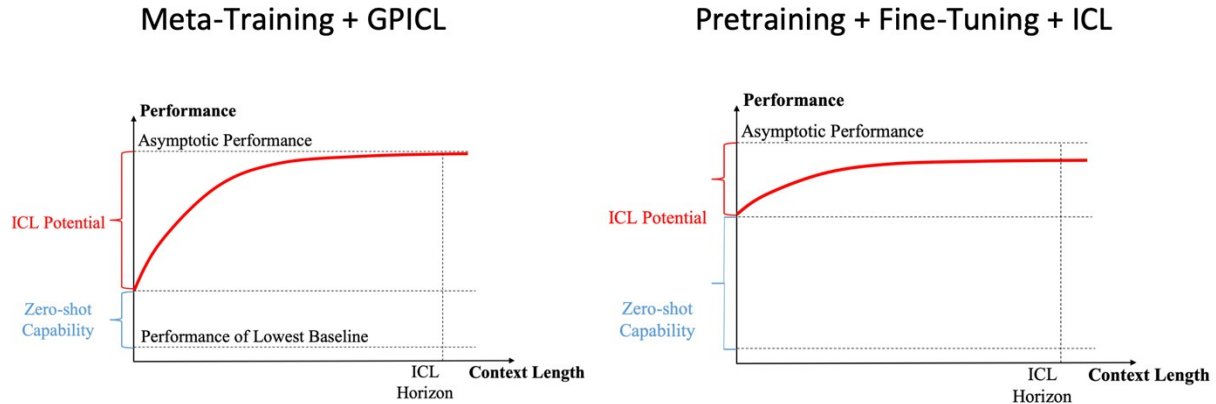


Figure 1. Foundation Model Options for Embodied AI [2]

Figure 1 compares two distinct approaches to foundational models for Embodied AI. The meta-training + GPICL approach, although exhibiting lower zero-shot performance and a smaller model size, excels in its ability to continuously learn from the environment. Over time, this enables the Embodied AI system to specialize in specific tasks. In contrast, the pretraining + fine-tuning + ICL approach, characterized by a larger model size and smaller context windows, delivers superior zero-shot performance but lacks the same level of continuous learning capability.

This distinction is supported by empirical evidence from the GPT-3 paper, which shows that a 7B Few-Shot model outperforms a 175B Zero-Shot model in terms of performance. Furthermore, if few-shot learning is replaced with a larger context window that enables the system to learn from its environment, performance could be further improved.

We propose that an ideal foundational model for Embodied AI should meet several key criteria. First, it should be capable of learning from complex instructions, demonstrations, and feedback without the need for manually crafted optimization techniques. Second, it must demonstrate high sample efficiency during the learning and adaptation processes. Third, the model should be able to continuously learn from contextual information, thereby avoiding catastrophic forgetting. Given these criteria, we conclude that the meta-learning + GPICL approach is more suited for Embodied AI systems. However, before committing to this approach, it is essential to examine the tradeoffs between these two strategies.

3. Computing Systems for Embodied AI

The development of Embodied AI is closely tied to the evolution of computing systems that can handle real-time decision-making and adaptation. These AI applications require robust computational infrastructures capable of processing complex sensory data, adapting models quickly, and allocating resources efficiently at scale [3, 4, 5]. However, there are three key

challenges that computing systems for Embodied AI must address: software stack complexity, limitations in computing architecture, and data bottlenecks.

Software Stack Complexity: Embodied AI necessitates a highly integrated software stack that enables seamless interactions between perception, cognition, and action modules. Unlike traditional AI systems that focus on single-task operations, Embodied AI demands a multi-layered stack that integrates perception algorithms with decision-making and physical actuation processes. This integration requires significant computational coordination and resource management to function efficiently.

Limitations in Computing Architecture: Traditional computing architectures are often inadequate for the specific demands of Embodied AI, where low-latency processing and parallel computing are crucial. To meet these demands, AI-specific hardware accelerators, such as dataflow architectures and neuromorphic chips, have been developed to address these limitations, offering better performance in real-time processing for autonomous systems.

Data Bottleneck and Efficiency: Embodied AI systems operate in dynamic, complex environments that generate vast amounts of diverse data. To overcome the data bottleneck, it is essential to implement both efficient storage solutions and real-time data processing techniques. Edge computing, which processes data closer to its source, is an effective strategy to mitigate data bottlenecks and improve response times, enabling faster decision-making in Embodied AI applications.

To address these challenges, we propose a layered software architecture that provides the necessary abstractions to manage software complexity while enabling flexibility:

1. **Control Adaptation Layer:** The control adaptation layer acts as an interface between core software and hardware components, simplifying the integration of various sensors, actuators, and control systems. It abstracts low-level complexities, allowing developers to focus on behavior logic and decision-making algorithms, thereby improving development efficiency.
2. **Core Robotic Functions:** Sitting atop the control adaptation layer, the core robotic function library provides essential packages for robot operations, such as mobility and user interaction. This layer offers developers high-level interfaces and ensures compatibility across different hardware platforms, significantly streamlining the development process.
3. **Robotic Applications:** The application layer enables the creation of sophisticated intelligent applications. Leveraging large foundational models, developers can integrate advanced AI capabilities into robots, enhancing their ability to perceive, understand, and interact with their environment.

To ensure the computational efficiency required for Embodied AI applications, a new architecture is necessary. This architecture must integrate multimodal sensors, offer efficient support for core robotic functions, and enable real-time processing for model-based robotic applications. By

addressing these needs, the proposed architecture will provide the foundation for next-generation Embodied AI systems capable of operating seamlessly in complex environments.

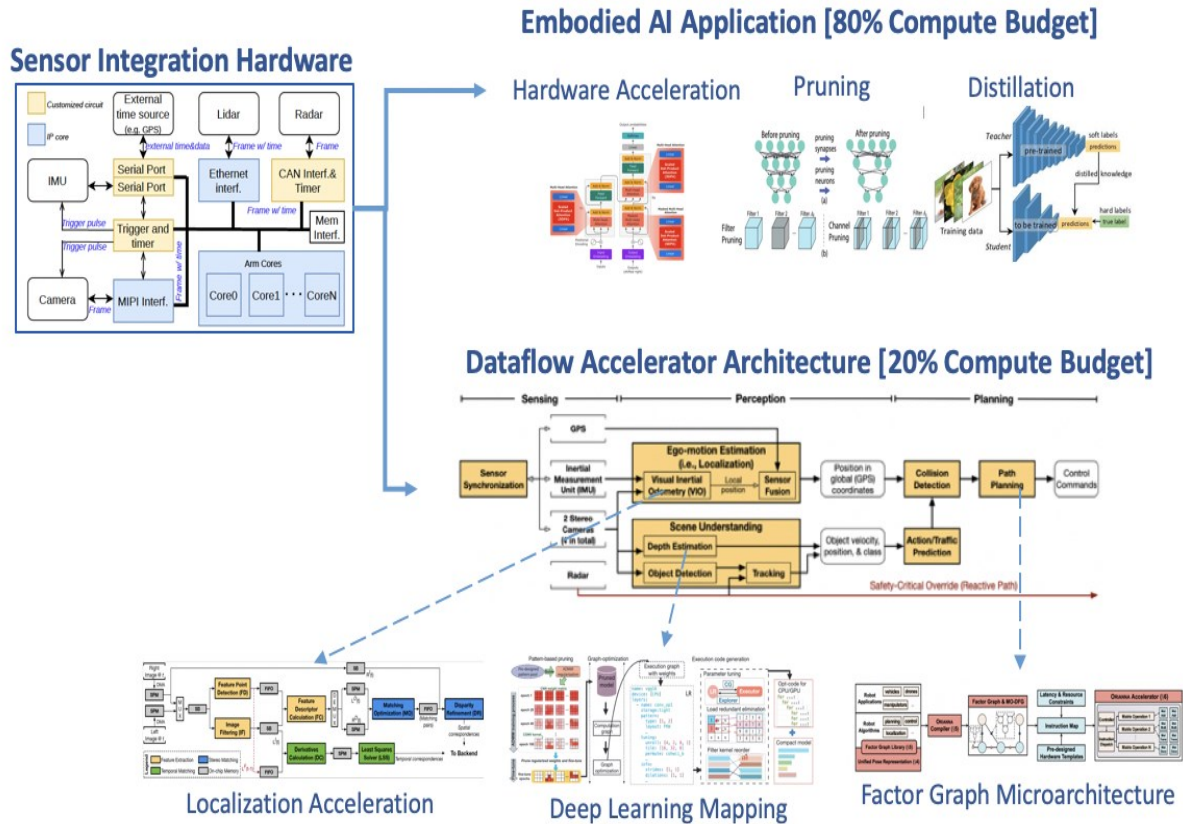


Figure 2. Computer Architecture for Embodied AI [2]

Sensor Integration and Synchronization: The effective integration and synchronization of sensors is paramount, particularly in robots equipped with multiple multimodal sensors. The quality of sensor integration directly influences the accuracy and reliability of the sensor data. To ensure precise data acquisition and integration, the hardware module must support the integration of over ten sensors and provide a unified timing source. This synchronization ensures the alignment of data streams, enabling seamless processing and accurate perception of the robot's environment.

Dataflow Accelerator Architecture: Robotic systems typically rely on the dataflow paradigm for computational tasks. To optimize computational efficiency in Embodied AI, it is essential to implement a Dataflow Accelerator Architecture that accelerates core robotic functions [6,7]. Our objective is to design an architecture that allocates no more than 20% of the total system's computational power to robotic core functions, preserving ample computational capacity for more complex and resource-intensive Embodied AI applications.

AI Agent Hardware Accelerator: Each Embodied AI application operates as an independent AI agent, responsible for understanding and executing complex tasks, often through visual language models (VLMs) or visual language action models (VLAMs). A primary challenge lies in executing these computationally intensive models, which can contain billions of parameters, in real-time on edge computing platforms. On the hardware side, the key to enhancing processing power lies in self-attentive hardware acceleration, which optimizes the processing of high-dimensional, diverse data inputs. On the software optimization side, techniques such as pruning and quantization can reduce model sizes by over 80% with minimal loss in accuracy, while knowledge distillation can further shrink model sizes by more than 90%. With these combined optimizations, we aim to enable the real-time processing of 70B-parameter models on edge platforms similar to the NVIDIA Jetson AGX Xavier, thus enhancing the responsiveness and intelligence of robotic applications within the constraints of edge computing resources.

4. The Value of Data in Embodied AI

Data serves as a crucial tool for monetization in both the internet and robotics sectors. Here, we explore the strategic value of data in Embodied AI, drawing on the internet sector as a historical benchmark [8].

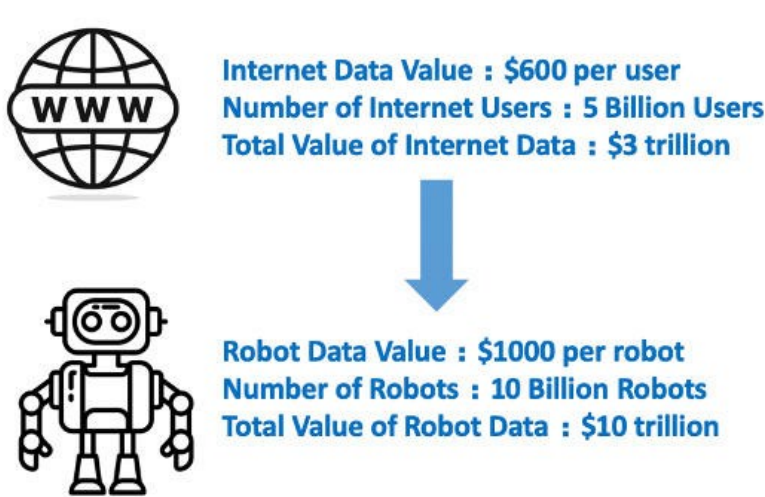


Figure 3. Data Monetization—Internet vs. Embodied AI [8]

In the internet industry, companies primarily capitalize on user data for targeted advertising and personalized content delivery. This tailored approach not only drives sales but also increases user engagement, potentially leading to higher subscription fees and more frequent usage. In contrast, in the Embodied AI sector, data plays a critical role in training deep learning models that enhance and optimize the capabilities of robotic systems.

Financially, user data holds substantial value for internet companies, with each user contributing an estimated \$600 in data value. With nearly 5 billion internet users globally, this equates to a total market value of approximately \$3 trillion for internet data.

Looking ahead to the Embodied AI sector, Elon Musk has forecasted a future where robots outnumber humans. Assuming market saturation with over 10 billion robots and considering the estimated cost of \$35,000 per robot post-commercialization, we conservatively estimate that robotic companies would allocate around 3% of the robot's cost towards data collection and generation. This investment would be aimed at advancing Embodied AI capabilities, leading to an estimated market value of Embodied AI data surpassing \$10 trillion—three times the value of internet data.

This projection underscores the enormous potential of data within the Embodied AI sector, though it's important to note that the industry is still in its early stages when it comes to data collection and generation.

While the future of the Embodied AI data industry is promising, current scalability is hindered by significant data bottlenecks. Unlike internet data, which consists primarily of user-generated inputs that are relatively simple to collect and aggregate, Embodied AI data arises from complex interactions between robots and their dynamic environments. While internet data is mined from user activities across digital platforms, Embodied AI data must capture a wide variety of physical interactions within unpredictable real-world environments.

For instance, while chat data from platforms like ChatGPT can be utilized to train models with 570 GB of text data, enabling impressive conversational abilities, training an Embodied AI system demands far more complex and multimodal data. This data, involving various sensory inputs and interaction types, is not only difficult to obtain but also costly to collect.

The first significant challenge in training Embodied AI systems is the need for large-scale, high-quality, and diverse datasets. For example, autonomous navigation robots require a vast array of environmental data to optimize their path planning and obstacle avoidance capabilities. Moreover, the accuracy of this data directly affects the performance of robots. Industrial robots performing high-precision tasks, for instance, need highly accurate data, as even minor errors can result in significant production issues. Additionally, the robot's ability to generalize across different environments depends on the diversity of data it processes. Home service robots, for example, must be able to adapt to various home environments and tasks, requiring them to learn from a broad spectrum of data to improve their generalization skills.

The second challenge is the issue of "data silos." Obtaining the comprehensive data required to train Embodied AI systems presents numerous challenges, including high costs, long timelines, and potential security risks. Most Embodied AI companies are limited to collecting data in controlled, specific environments, which exacerbates the problem. The lack of collaboration and data sharing among organizations further compounds the situation, resulting in duplicated efforts, wasted resources, and the formation of "data silos." These silos significantly hinder progress within the Embodied AI field, preventing the development of more robust and adaptable systems.

5. Societal Impact of Embodied AI

The widespread adoption of Embodied AI holds transformative potential across multiple sectors, including healthcare, manufacturing, and service industries. However, these advancements also bring complex societal implications, encompassing both opportunities and challenges [9,10,11].

Healthcare: Embodied AI-driven technologies can enhance patient care by enabling assistive robots to interact with patients in hospitals, rehabilitation centers, and homes. Robots equipped with Embodied AI can perform tasks like patient monitoring, medication dispensing, and mobility assistance, reducing the burden on healthcare providers. This capability also extends to remote healthcare services, where Embodied AI systems can provide medical assistance in areas with limited access to healthcare facilities.

Manufacturing and Industry: The manufacturing sector has embraced robotics for decades, but Embodied AI promises to take industrial automation to new heights. By integrating sensory perception and adaptive learning, Embodied AI-powered robots can optimize production lines, respond to environmental changes, and collaborate more effectively with human workers. This adaptability makes Embodied AI an invaluable asset in complex manufacturing environments where human-robot collaboration is critical.

Service Industries: Embodied AI is set to revolutionize the service industry by enabling robots to perform customer-facing tasks in retail, hospitality, and transportation. For instance, robots equipped with Embodied AI can engage with customers in real time, answer inquiries, and adapt their interactions based on customer responses. Such systems could enhance customer experiences, providing a personalized touch in various service applications.

Ethical and Regulatory Considerations: The integration of Embodied AI into daily life raises ethical questions around privacy, autonomy, and accountability. As robots and AI systems become more autonomous, issues of safety, transparency, and human rights come to the forefront. Policymakers and technologists must work together to develop frameworks that ensure the responsible deployment of Embodied AI, protecting both individual rights and societal interests.

Future Directions: Embodied AI's impact on society will continue to grow, necessitating ongoing research into its societal and ethical implications. Research should address not only the technical and economic benefits of Embodied AI but also the long-term effects on human labor, social equity, and ethical governance. As Embodied AI evolves, cross-disciplinary collaboration will be essential in shaping its integration into society.

6. Conclusion

The potential of Embodied AI to reshape industries, enhance human-robot collaboration, and provide societal benefits is vast. By advancing foundational models, robust computing systems, and data-driven frameworks, Embodied AI is poised to unlock new levels of machine autonomy and adaptability. However, with these advancements come challenges, including data privacy

concerns, energy consumption of embodied AI systems, regulatory requirements, and ethical implications.

Future research in Embodied AI should focus on refining foundational models that support task generalization, developing hardware architectures optimized for real-time adaptation, and creating unified data standards to overcome data silos. Furthermore, as Embodied AI systems become more prevalent, ethical considerations around transparency, safety, and human-robot interaction will be paramount.

The continued development of Embodied AI represents an interdisciplinary effort, combining insights from artificial intelligence, robotics, cognitive science, and sociology. As we navigate this transformative journey, collaboration across sectors and a focus on responsible innovation will be critical in realizing the full potential of Embodied AI.

References

1. Liu, S., Wu, S., A Brief History of Embodied AI, and its Outlook, Communications of the ACM, <https://cacm.acm.org/blogcacm/a-brief-history-of-embodied-artificial-intelligence-and-its-future-outlook/>
2. Wang, F., Liu, S., Building Foundation Models for Embodied AI, Communications of the ACM, <https://cacm.acm.org/blogcacm/building-foundation-models-for-embodied-artificial-intelligence/>
3. Liu, S., Ding, N., Building Computing Systems for Embodied AI, Communications of the ACM, <https://cacm.acm.org/blogcacm/building-computing-systems-for-embodied-artificial-intelligence/>
4. Wan, Z., Gan, Y., Yu, B., Liu, S., Raychowdhury, A. and Zhu, Y., 2024. The Vulnerability-Adaptive Protection Paradigm. Communications of the ACM, 67(9), pp.66-77.
5. Liu, S., Liu, L., Tang, J., Yu, B., Wang, Y. and Shi, W., 2019. Edge computing for autonomous driving: Opportunities and challenges. Proceedings of the IEEE, 107(8).
6. Liu, S., Tang, J., Zhang, Z. and Gaudiot, J.L., 2017. Computer architectures for autonomous driving. Computer, 50(8)
7. Yu, B., Hu, W., Xu, L., Tang, J., Liu, S. and Zhu, Y., 2020, October. Building the computing system for autonomous micromobility vehicles: Design constraints and architectural optimizations. In 2020 53rd Annual IEEE/ACM International Symposium on Microarchitecture (MICRO)
8. Liu, S., The Value of Data in Embodied AI, Communications of the ACM, <https://cacm.acm.org/blogcacm/the-value-of-data-in-embodied-artificial-intelligence/>
9. Liu, S., Societal Impacts of Embodied AI, Communications of the ACM, <https://cacm.acm.org/blogcacm/societal-impacts-of-embodied-ai/>
10. Liu, S., 2024. Shaping the Outlook for the Autonomy Economy. Communications of the ACM, 67(6), pp.10-12.
11. Wu, S., Yu, B., Liu, S. and Zhu, Y., 2023. Autonomy 2.0: The Quest for Economies of Scale. arXiv preprint arXiv:2307.03973.
12. Brooks, R.A., 1991. Intelligence without representation. Artificial intelligence, 47(1-3), pp.139-159.
13. Pfeifer, R. and Scheier, C., 2001. Understanding intelligence. MIT press.
14. Smith, L.B., 2005. Cognition as a dynamic system: Principles from embodiment. Developmental Review, 25(3-4), pp.278-298.

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